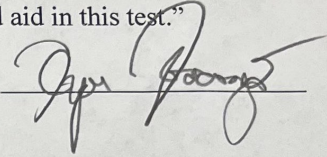


Final Exam
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- Make sure you write your name.
- You can use a computer.
- This exam is open book and notes.
- To receive credit, please be neat and show all key steps of your solutions clearly.
- Don't forget to submit it to **Gradescope**.
- Please sign the following pledge before beginning the exam:

"I have neither given nor received any unauthorized aid in this test."

Signature: 

Problem #	Score
1	/35
2	/30
3	/20
4	/15
Total	/100
Comprehensive	(S/U)

Problem 1 (35 pts)

A 4R manipulator is shown in Figure 1. The nonzero link parameters are $\alpha_1 = -90^\circ$, $d_2 = 1$, $\alpha_2 = 45^\circ$, $d_3 = 1$, and $a_3 = 1$. The mechanism is pictured in the configuration corresponding to $\theta = [0, 0, 90^\circ, 0]^\top$.

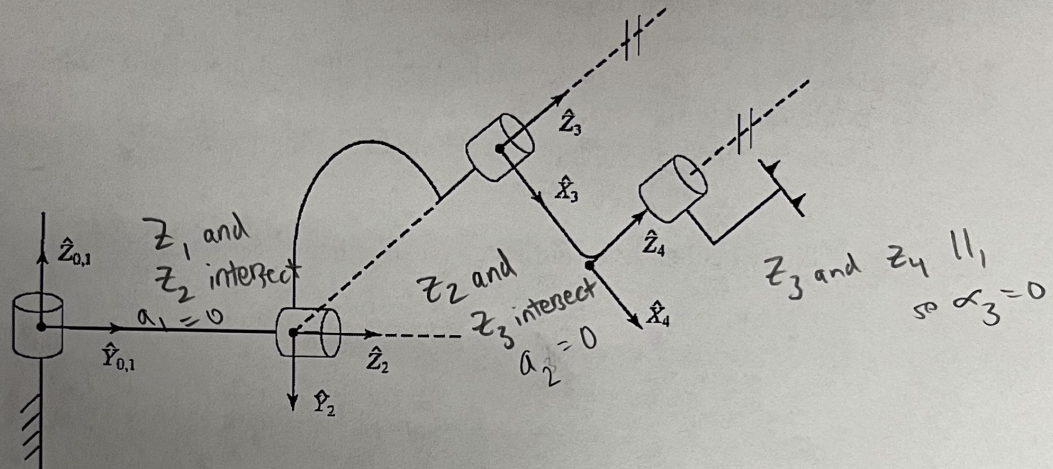


Figure 1: A 4R manipulator shown in the configuration $\theta = [0, 0, 90^\circ, 0]^\top$.

- Find ${}^0P_{4ORG}$ using forward kinematics. (20 pts)
- Derive the inverse kinematics for $[\theta_1, \theta_2, \theta_3]$ given ${}^0P_{4ORG} = [p_x, p_y, p_z]^\top$. (Hint: use Pieper's solution or try $p_x^2 + p_y^2 + p_z^2$ for θ_3 ; if you cannot get θ_3 , you can assume it's known and solve for θ_1 and θ_2 for partial credit; 10 pts)
- Find all values of $[\theta_1, \theta_2, \theta_3]$ such that ${}^0P_{4ORG} = [0, 1, \sqrt{2}]^\top$ and ${}^0P_{4ORG} = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]^\top$. (You are **NOT** required to use the inverse kinematics; 2.5 pts + 2.5 pts)

1) DH Table

$i-1$	i	a_{i-1}	α_{i-1}	d_i	θ_i
0	1	0	0	0	θ_1
1	2	0	-90°	1	θ_2
2	3	0	45°	1	θ_3
3	4	1	0	0	θ_4

$${}^{i-1}T_i = \begin{pmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^0_4T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T$$

$${}^0_1T = \begin{pmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2T = \begin{pmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} c_3 & -s_3 & 0 & 0 \\ s_3 \frac{\sqrt{2}}{2} & c_3 \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ s_3 \frac{\sqrt{2}}{2} & c_3 \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3_4T = \begin{pmatrix} c_4 & -s_4 & 0 & 1 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

→ multiply m's

$${}^0_4T = \begin{pmatrix} 0 & R & 0 & p_{4org} \\ 4 & & & \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{by definition of } T)$$

→ MATLAB

$${}^0P_{4org} = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} \cos \theta_1 (\frac{\sqrt{2}}{2} \sin \theta_2 + \cos \theta_2 \cos \theta_3 - \frac{\sqrt{2}}{2} \sin \theta_2 \sin \theta_3) + \sin \theta_1 (-\frac{\sqrt{2}}{2} - 1 - \frac{\sqrt{2}}{2} \sin \theta_3) \\ \cos \theta_1 (1 + \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \sin \theta_3) + \sin \theta_1 (\frac{\sqrt{2}}{2} \sin \theta_2 + \cos \theta_2 \cos \theta_3 - \frac{\sqrt{2}}{2} \sin \theta_2 \sin \theta_3) \\ -\cos \theta_3 \sin \theta_2 + \frac{\sqrt{2}}{2} \cos \theta_2 (1 - \sin \theta_3) \end{pmatrix}$$

2) IK $p_x^2 + p_y^2 + p_z^2 \rightarrow$ MATLAB simplify (expand($p_x^2 + p_y^2 + p_z^2$)) = $\sqrt{2} + 3 + \sqrt{2} \sin \theta_3$

Solve θ_3

$$\frac{p_x^2 + p_y^2 + p_z^2 - \sqrt{2} - 3}{\sqrt{2}} = \sin \theta_3 \rightarrow \cos \theta_3 = \pm \sqrt{1 - \sin^2 \theta_3}$$

Use Atan2

$$\theta_3 = \text{Atan2} \left(\frac{p_x^2 + p_y^2 + p_z^2 - \sqrt{2} - 3}{\sqrt{2}}, \pm \sqrt{1 - \sin^2 \theta_3} \right)$$

Solve θ_2 Use $p_z = -c_3 s_2 + \frac{\sqrt{2}}{2} c_2 (1 - s_3)$, where θ_3 is now known. Only var is θ_2 .

Use form $a \cos \theta_2 + b \sin \theta_2 = c$, where $a = \frac{\sqrt{2}}{2} (1 - s_3)$, $b = -c_3$, $c = z$

$$\theta_2 = \text{Atan2} \left(\pm \sqrt{a^2 + b^2 - z^2}, z \right) + \text{Atan2}(b, a)$$

Solve θ_1

look at p_x and p_y . They are in the form $A \cos \theta_1 + C \sin \theta_1 = d = p_x$
 $E \cos \theta_1 + F \sin \theta_1 = g = p_y$

b/c θ_2
and θ_3 are
now known.

$$\theta_1 = \text{atan2}(A g - d E, d F - g)$$

From p_x , $A = \frac{\sqrt{2}}{2} \sin \theta_2 (1 - \sin \theta_3) + \cos \theta_2 \cos \theta_3$
 $C = -(\frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} \sin \theta_3)$

By inspection, $A = F$ and $C = -E$

→ Therefore, $\theta_1 = \text{Atan2}(A p_y + p_x C, p_x A - C p_y)$

3) $P_x = 0$
 $P_y = 1 \rightarrow S_3 = \frac{1 + 2 - \sqrt{2} - 3}{\sqrt{2}} = -1$ Final Exam/MAE C163A/C263A/Fall 2022/DWH
 $P_z = \sqrt{2} \rightarrow C_3 = \sqrt{1 - 1} = 0$

$$\theta_3 = \text{Atan2}(-1, 0) = -90$$

For θ_2 :

$$a = \sqrt{2} \quad b = 0 \quad z = \sqrt{2} \rightarrow \sqrt{a^2 + b^2 - z^2} = 0$$

$$\theta_2 = \text{Atan2}(0, \sqrt{2}) + \text{Atan2}(0, \sqrt{2}) \Rightarrow \theta_2 = 0$$

$$\theta_1 = \text{Atan2}(A, -C), \text{ b/c } x=0 \text{ and } y=1$$

$$= \text{Atan2}(0, +1) \rightarrow \theta_1 = 0$$

$$P_{\text{long}} = \begin{pmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{pmatrix}$$

$$\text{Find } \theta_3 \rightarrow \text{Atan2}\left(\frac{\frac{1}{2} + \frac{1}{2} - \sqrt{2} - 3}{\sqrt{2}}, \pm \sqrt{1 - \sin^2 \theta_3}\right) \rightarrow \text{gives mixed answer}$$

$$\begin{pmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \\ 0 \end{pmatrix} \text{ not in workspace}$$

Problem 2 (30 pts)

Imagine an arm like PUMA 560, except that joint 3 is replaced with a prismatic joint. Assume the prismatic joint slides along the direction of Z_3 shown in Figure 2. However, there is still an offset equivalent to d_3 to be accounted for. The schematic of the robot wrist is shown in Figure 3. The origins of frame 4, 5 and 6 are all at the same point.

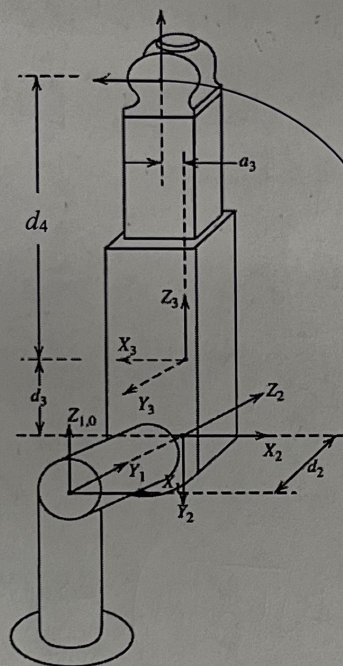


Figure 2: Frame assignments of the manipulator.

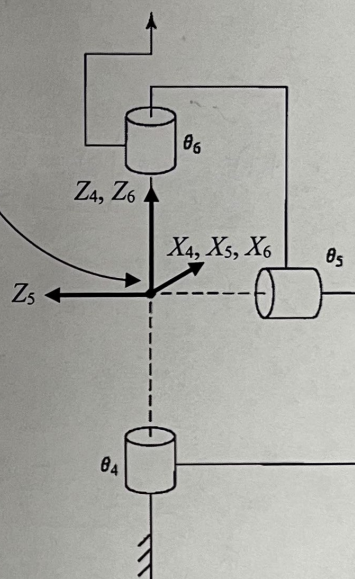


Figure 3: Wrist schematic.

1. Complete the DH table. (10 pts)

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	-90	0	d_2	θ_2
3	90	0	d_3	180
4	0	a_3	d_4	θ_4
5	-90	0	0	θ_5
6	90	0	0	θ_6

2. Write out all the individual transformation matrices between joints and describe how to calculate the overall transformation matrix from frame 0 to frame 6. (You do **NOT** need to calculate 0_6T ; 10 pts)

3. Given ${}^4_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, solve for θ_5 and θ_6 symbolically. (10 pts)

$${}^0_6T = \begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 \\ \begin{matrix} 0 \\ 6 \end{matrix} & T & T & T & T & T & T \\ & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$$

$${}^0_1T = \begin{pmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^1_2T = \begin{pmatrix} c_2 & -s_2 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^2_3T = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^3_4T = \begin{pmatrix} c_4 & -s_4 & 0 & a_3 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^4_5T = \begin{pmatrix} c_5 & -s_5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_5 & -c_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^5_6T = \begin{pmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_6 & c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$${}^4_6T = \begin{pmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \theta_5 \text{ and } \theta_6$$

Matlab to get ${}^4_6T = {}^4_5T {}^5_6T$

$$= \begin{pmatrix} c_5 c_6 & -c_5 s_6 & s_5 & 0 \\ s_6 & c_6 & 0 & 0 \\ -c_6 s_5 & s_5 s_6 & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

θ_5 , use r_{13} and r_{33}

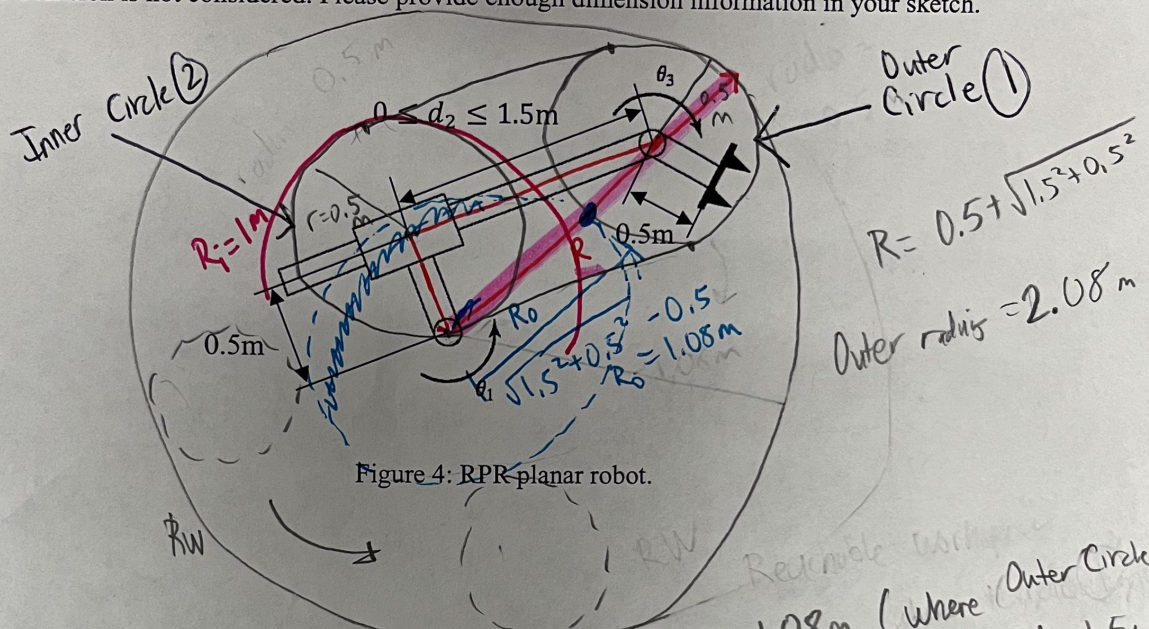
$$\theta_5 = \text{Atan2}(r_{13}, r_{33})$$

θ_6 , use r_{21} and r_{22}

$$\theta_6 = \text{Atan2}(r_{21}, r_{22})$$

Problem 3 (20 pts)

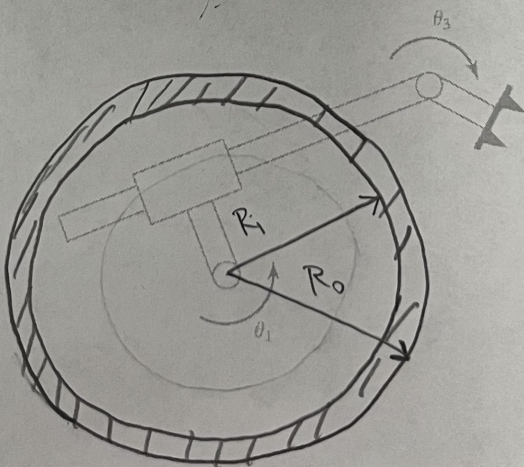
Sketch the **dexterous** workspace of the RPR planar robot. All revolute joints can rotate 360° and self-collision is not considered. Please provide enough dimension information in your sketch.



• $R_{o, \text{outer}}$ limit of DW is @ $r = 1.08\text{m}$ (where Outer Circle ①, when $d_2 = 1.5$, cannot touch)

- inner limit, when $d_2 = 0$, gives R_i of 1 (Inner Circle ② cannot touch)

DW is from $R_i = 1\text{m}$ to $R_o = 1.08\text{m}$



Problem 4 (15 pts)

In Figure 5, two 3R mechanisms are pictured. In both cases, the three axes intersect at a point. The mechanism of (a) has link twists of magnitude 90 degrees. The mechanism of (b) has one twist of ϕ ($< 90^\circ$) in magnitude and the other of $180 - \phi$ in magnitude. The mechanism of (a) can be seen to be in correspondence with Z-Y-Z Euler angles, and therefore we know that it suffices to orient link 3 (with arrow in figure) arbitrarily with respect to the link 0 (base). In other words, the arrow can point to the intersection in any direction. Because ϕ is not equal to 90 degrees, it turns out that the other mechanism cannot orient link 3 arbitrarily.

Sketch the set of orientations that are **unreachable** with the mechanism of (b). Please provide enough dimension information. Note that we assume that all joints can turn 360 degrees (i.e., no limits) and the links may pass through each other if need be (i.e., workspace not limited by self-collision).

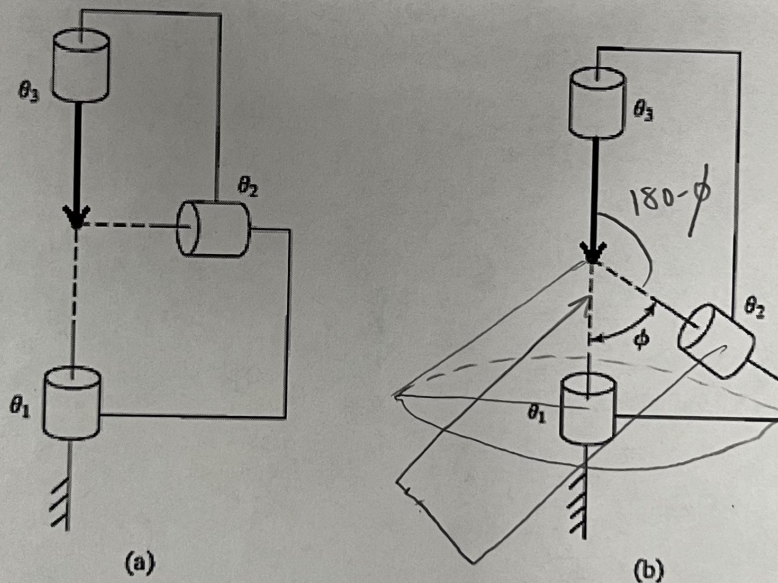
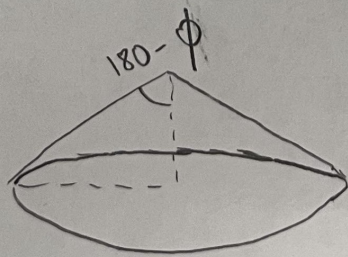


Figure 5: Two 3R mechanisms.

Arrow cannot reach cone:



• θ_3 rotation creates a symmetric unreachable shape

• θ_1 and θ_2 sweep out a cone